

GR&R Study Report

Gauge Repeatability & Reproducibility — AIAG MSA 4th Edition

Equipment / Gage:	Vernier Caliper VC-07 SN-1138
Study Performed By:	QE Lab - Example B
Report Date:	2026-05-10
Study Design:	3 Operators, AIAG Crossed GR&R
Regulatory Reference:	21 CFR 820.72 AIAG MSA 4th Ed.

Variation Components Summary

Component	Study Variation	% of TV	Threshold
EV — Equipment (Repeatability)	0.0381	60.1%	—
AV — Appraiser (Reproducibility)	0.0424	66.9%	—
GR&R — Combined	0.0570	89.9%	≤ 10%: Accept ≤ 30%: Marginal
PV — Part Variation	0.0277	43.7%	—
TV — Total Variation	0.0634	100.0%	—
ndc — Number of Distinct Categories	0.7	—	≥ 5 required

Intermediate Calculation Values

Parameter	Symbol	Value	Formula
Grand Mean	X-bar-bar	24.9956	Mean of all measurements
Avg Range (overall)	R-bar-bar	0.0125	Mean of operator R-bars
Operator Mean Spread	X-diff	0.0159	max(X-bar op) - min(X-bar op)
Part Range	Rp	0.0171	max(part avg) - min(part avg)

Per-Operator Breakdown

Operator	Grand Mean (X-bar)	Avg Range (R-bar)
Alice	24.9950	0.0131
Bob	25.0039	0.0114
Carol	24.9880	0.0130

Acceptance Determination

[FAIL] UNACCEPTABLE

%GR&R = 89.9% | ndc = 0.7

The measurement system is UNACCEPTABLE. %GR&R; exceeds 30%. Do not use this gage for product acceptance. Identify and correct root cause before re-study. Common causes: gage discrimination, worn components, inconsistent measurement technique, environmental factors, or inadequate operator training.

ndc = 0.7 — INADEQUATE: ndc < 5. The gage cannot resolve enough part categories for meaningful process control. Gage improvement is required.

AIAG Constants Applied — K1=? (10 trials), K2=2.7 (3 operators), K3 per part count. Study variation = 5.15σ (99% of normal distribution).

Reference: AIAG Measurement Systems Analysis Reference Manual, 4th Edition (2010). Regulatory basis: 21 CFR Part 820.72 — Inspection, Measuring, and Test Equipment (FDA Quality System Regulation). This report is a controlled quality record. Retain per applicable document control procedures.